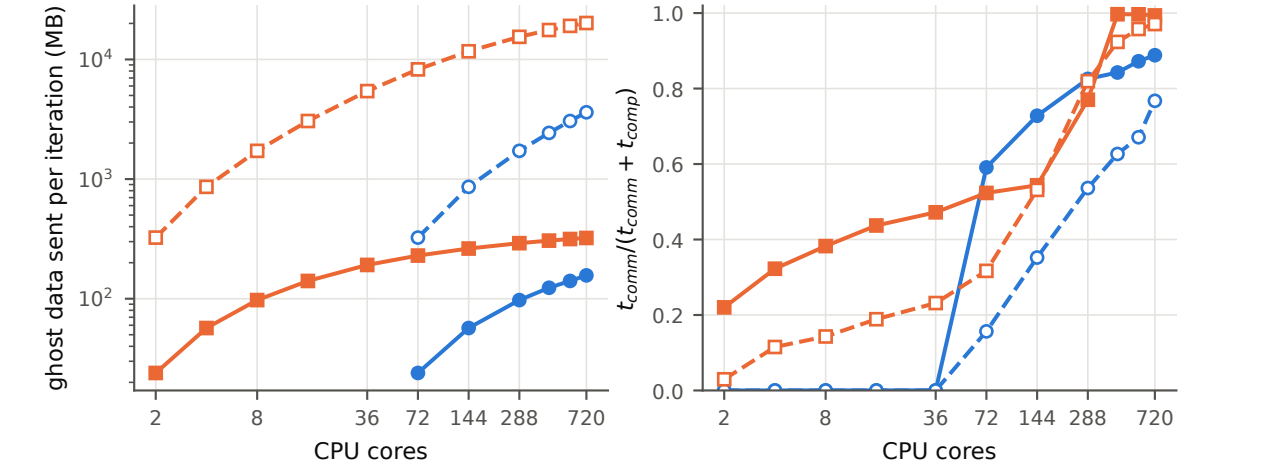
(a) Ghost data per iteration
ghost data sent per iteration (MB)
10^4
10^3
10^2
2
8
36
72
144
288
720
CPU cores
(b) Share of time in MPI calls
$t_{comm}/(t_{comm} + t_{comp})$
1.0
0.8
0.6
0.4
0.2
0.0
2
8
36
72
144
288
720
CPU cores
MPI+OpenMP, soc-LiveJournal1
pure MPI, soc-LiveJournal1
MPI+OpenMP, com-friendster
pure MPI, com-friendster